Arista Access Node WHI-ACC-61 Whitefield East POP Optical Transport Failure SOP

Node: WHI-ACC-61 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_8707

Description

The Access Node WHI-ACC-61 at the Whitefield East POP is equipped with Arista switches, specifically designed to handle the high volumes of traffic associated with modern telecommunications networks. The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) in this specific role and vendor equipment is primarily attributed to the inherent fragility of optical interfaces, which can be prone to degradation over time due to factors such as environmental stress, manufacturing defects, or physical damage. Furthermore, the role of the Access Node as a critical entry point for traffic into the network makes it particularly susceptible to issues related to optical transport. The underlying protocol mechanics involved in this failure domain include the SFP/XFP's ability to transmit and receive optical signals, which can be disrupted by degradation or physical damage to the interface. In the context of Arista equipment, the CLI dialect plays a crucial role in diagnosing and resolving such issues. Through the judicious use of Arista-specific commands, engineers can gain valuable insights into the status of the optical interface, identify potential root causes, and implement targeted fixes to restore service.

Pre-requisite Checks

- Verify that the Whitefield East POP is accessible and that all necessary personnel are present for remote hands support.

- Check the status of redundant links to ensure that there are no other points of failure that could impact the resolution process.

- Ensure that all necessary network diagrams and documentation are readily available for reference.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces status | i WHI-ACC-61, Expected Output: WHI-ACC-61/1/1/1: SFP-10G-SR up, Description: This command will display the status of the interface WHI-ACC-61/1/1/1. The expected output indicates that the interface is up and operational.
```

```
Command: show interfaces transceivers | i WHI-ACC-61, Expected Output:
WHI-ACC-61/1/1/1: Transceiver: SFP-10G-SR, Status: OK, Description:
This command will display the status of the transceiver on interface
WHI-ACC-61/1/1/1. The expected output indicates that the transceiver
is functioning properly.

Command: show logging | i WHI-ACC-61, Expected Output: Jun 12 14:30:00
WHI-ACC-61/1/1/1: %LINK-3-UPDOWN: Interface WHI-ACC-61/1/1/1, changed
state to down, Description: This command will display the logging
information for interface WHI-ACC-61/1/1/1. The expected output
indicates that the interface has gone down, which may be related to
the optical transport failure.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP, Command: replace /interfaces transceivers /1/1/1 SFP-10G-SR, Description: This step involves replacing the SFP/XFP with a new one. The command above will replace the SFP/XFP on interface WHI-ACC-61/1/1/1 with a new SFP-10G-SR.

Step 2: Step: Verify the fiber connection, Command: show interfaces status | i WHI-ACC-61, Description: This step involves verifying the fiber connection to ensure that it is properly connected and functioning. The command above will display the status of the interface WHI-ACC-61/1/1/1.

Post-Check Verification

```
Command: show interfaces status | i WHI-ACC-61, Expected Output:
WHI-ACC-61/1/1/1: SFP-10G-SR up, Description: This command will verify
that the interface WHI-ACC-61/1/1/1 is up and operational.

Command: show interfaces transceivers | i WHI-ACC-61, Expected Output:
WHI-ACC-61/1/1/1: Transceiver: SFP-10G-SR, Status: OK, Description:
This command will verify that the transceiver on interface
WHI-ACC-61/1/1/1 is functioning properly.
```

Roll-back Procedures

- Step: Revert to original SFP/XFP, Command: replace /interfaces transceivers /1/1/1 original-SFP-10G-SR, Description: This step involves reverting the SFP/XFP to its original state. The command above will replace the SFP/XFP on interface WHI-ACC-61/1/1/1 with the original SFP-10G-SR.

- Step: Verify the fiber connection, Command: show interfaces status | i WHI-ACC-61, Description: This step involves verifying the fiber connection to ensure that it is properly connected and functioning. The command above will display the status of the interface WHI-ACC-61/1/1/1.